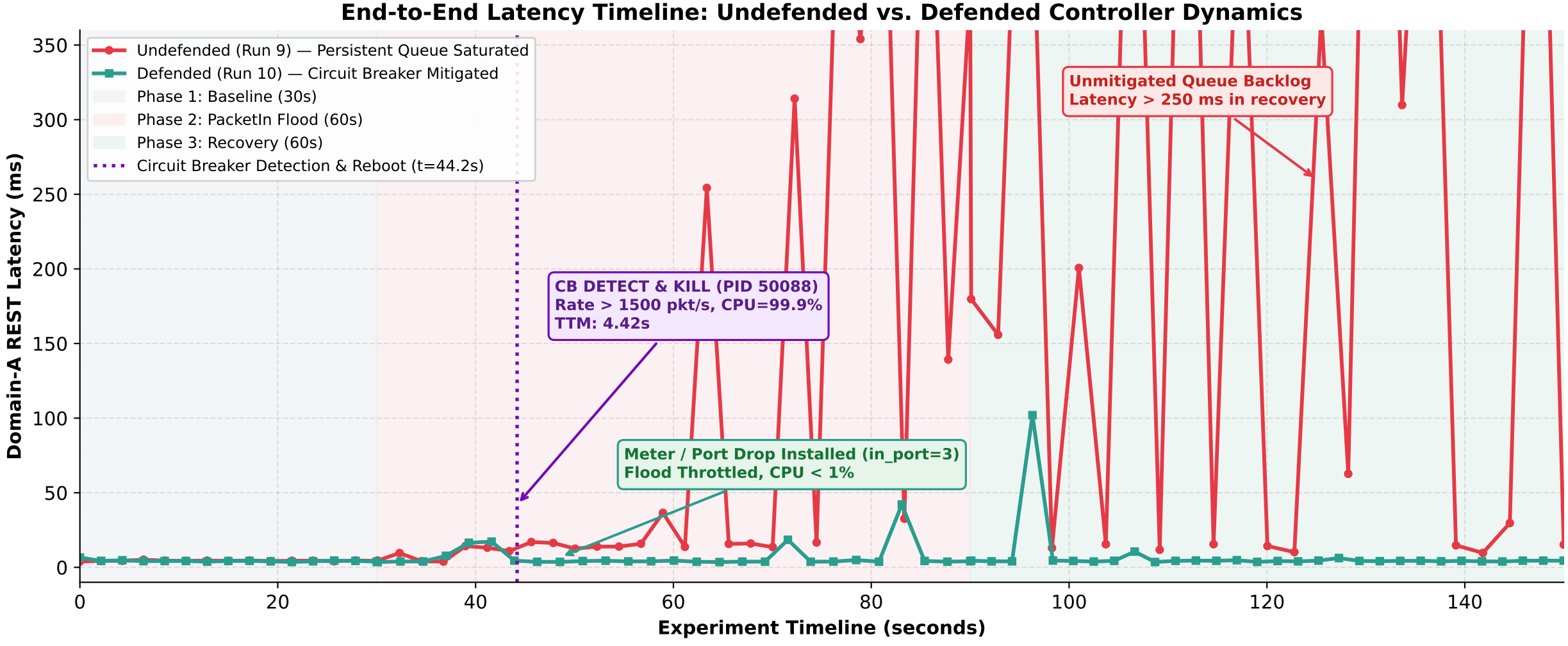
End-to-End Latency Timeline: Undefended vs. Defended Controller Dynamics
Undefended (Run 9) — Persistent Queue Saturated
Defended (Run 10) — Circuit Breaker Mitigated
Phase 1: Baseline (30s)
Phase 2: PacketIn Flood (60s)
Phase 3: Recovery (60s)
Circuit Breaker Detection & Reboot (t=44.2s)
CB DETECT & KILL (PID 50088)
Rate > 1500 pkt/s, CPU=99.9%
TTM: 4.42s
Meter / Port Drop Installed (in_port=3)
Flood Throttled, CPU < 1%
Unmitigated Queue Backlog
Latency > 250 ms in recovery
Domain-A REST Latency (ms)
Experiment Timeline (seconds)
0
50
100
150
200
250
300
350
0
20
40
60
80
100
120
140